Free balancing equations worksheet worksheet 25

Maths | 10th grade | Algebra

1. If $3(x - 2) + 5 = 2x + 9$, what is the value of x?

A) 1
B) 2
C) 3
D) 4

2. Solve for y: $4 - 2y = 2(y + 5)$.

A) 0
B) -2
C) 5
D) 2

3. For what value of k does $x = k$ not satisfy the equation $2x^2 - 3x = k + 5$?

A) 2
B) 5
C) 3
D) 0

4. Find the value of m such that the equation $7 - m(x - 4) = 9$ is satisfied for $x = 2$.

A) 0.5
B) 1
C) 2
D) 8

5. What is the value of n for which $\frac{n - 2}{3} + \frac{3n}{2} = 5$?

A) 4
B) 6
C) 8
D) 10

6. Determine a if $\sqrt{3a + 2} - \sqrt{a} = 2$.

A) 4
B) 9
C) 1
D) 2

7. Find the value of t such that $(t - 1)^2 = t + 3$.

A) 0
B) 5
C) 3
D) 2

8. If $f(x) = 2x^2 - 4x + 1$, what is $f(x + 1)$?

A) $2x^2$
B) $2x^2 + 2x - 1$

C) $2x^2 - 2x + 3$
D) $2x^2 + 6x + 1$

9. Find x such that $\frac{x+3}{x-3} = \frac{5}{2}$.

A) 7
B) 8
C) -3
D) 5

10. If $ax + 6 = 3x + 2a$ and $x = 4$, find a.

A) 2
B) 4
C) 7
D) 1

Answer Key

1. D
Distribute and simplify: $3x - 6 + 5 = 2x + 9$ which simplifies to $3x - 1 = 2x + 9$. Solve for x: $x = 10$.

2. B
Expand and simplify: $4 - 2y = 2y + 10$, leading to $-4y = 6$, so $y = -1.5$.

3. C
Substitute and solve $2k^2 - 3k = k + 5$. Solving gives $2k^2 - 4k - 5 = 0$. Using the quadratic formula, solutions are $k = 2.5, -1$. For $k=3$, no solution exists.

4. A
Substitute $x = 2$ into the equation: $7 - m(-2) = 9$, then $7 + 2m = 9$, so $m = 1$.

5. A
Clear fractions by multiplying through by 6: $2(n-2) + 9n = 30$. Simplifying gives $11n - 4 = 30$, so $11n = 34$, thus $n = 34/11$.

6. B
Square both sides: $3a + 2 = (\sqrt{a} + 2)^2$. Simplifying gives $3a + 2 = a + 4\sqrt{a} + 4$. Set rational parts equal and solve: $2a - 2 = 4\sqrt{a}$, leads to $a = 9$.

7. C
Expand: $t^2 - 2t + 1 = t + 3$, which simplifies to $t^2 - 3t - 2 = 0$. Solving gives $t = 3$ and $t = -1$. $t = 3$ satisfies the original equation.

8. C
Substitute: $f(x+1) = 2(x+1)^2 - 4(x+1) + 1$. This expands to $2x^2 + 4x + 2 - 4x - 4 + 1$, simplifying to $2x^2 - 2x + 3$.

9. B
Cross-multiply: $2(x + 3) = 5(x - 3)$, which gives $2x + 6 = 5x - 15$. Solving gives $x = 7$.

10. D
Substitute $x = 4$: $4a + 6 = 12 + 2a$. Simplifying gives $2a + 6 = 12$, so $2a = 6$, therefore $a = 3$.